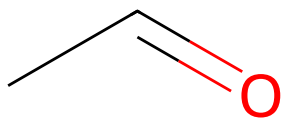


Chemistry — Assignment 5: IUPAC Nomenclature

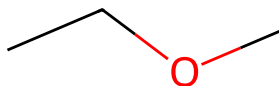
Anshuman Atrey · 150096724029 · BTech CSE 2024-28

Chemistry (Foundational Chemistry) · Semester 5

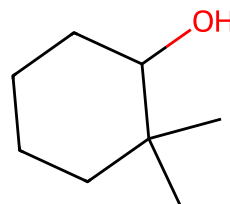
Question. Name / draw / give rules for the IUPAC set (structures and short answers below).



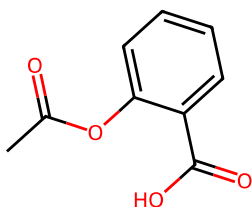
$\text{CH}_3\text{CHO} \rightarrow$ **ethanal**



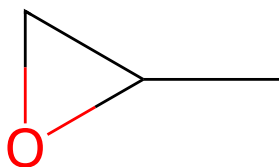
$\text{CH}_3\text{-O-C}_2\text{H}_5 \rightarrow$ **methoxyethane**



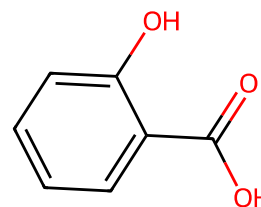
3,3-dimethylcyclohexan-1-ol $\rightarrow$ **3,3-dimethylcyclohexan-1-ol**



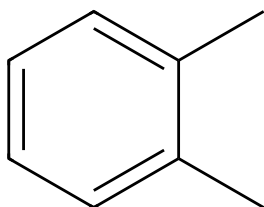
acetylsalicylic acid $\rightarrow$ **2-acetoxybenzoic acid**



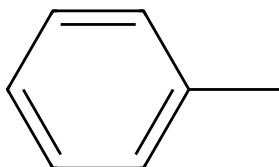
propylene oxide $\rightarrow$ **2-methyloxirane**



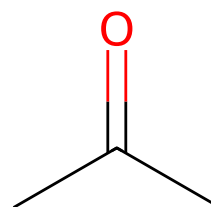
salicylic acid $\rightarrow$ **2-hydroxybenzoic acid**



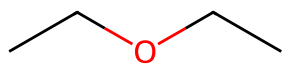
o-xylene $\rightarrow$ **1,2-dimethylbenzene**



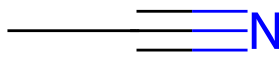
toluene $\rightarrow$ **methylbenzene**



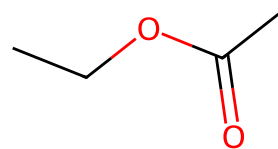
acetone $\rightarrow$ **propan-2-one**



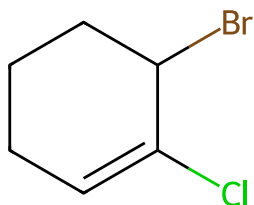
diethyl ether → **ethoxyethane**



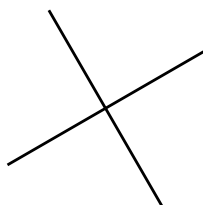
acetonitrile → **ethanenitrile**



ethyl acetate → **ethyl ethanoate**



3-bromo-1-chlorocyclohexene → **3-bromo-1-chlorocyclohexene**



neopentane → **2,2-dimethylpropane**

Short answers

- Formula of ethanol: **CH₃CH₂OH (C₂H₅OH)**
- Smallest alkane that can form a ring: **cyclopropane (C₃H₆)**
- Dienes: **cumulated** (C=C=C adjacent), **conjugated** (alternating C=C–C=C), **isolated** (separated by >1 sp³ C)
- What is a functional group: **the atom/group that determines a molecule's characteristic reactions**
- MCQ (neopentane): **2,2-dimethylpropane**; MCQ (heptane): **3-ethyl-4,4-dimethylheptane**
- 4-bromo-3-methylpent-2-ene; 4-propyl-3,5-dimethylhept-1-en-6-yne (names as given)

Order of priority of functional groups

carboxylic acid > sulfonic acid > ester > acid halide > amide > nitrile > aldehyde > ketone > alcohol > amine > alkene > alkyne > alkane

Rules of IUPAC nomenclature

1. Select the longest chain containing the highest-priority functional group (the parent).
2. Change the parent alkane/ene/yne ending to the suffix of that group.
3. Number the chain from the end nearest the highest-priority group (lowest locant).
4. Name the remaining groups as prefixes in alphabetical order.
5. Assign the lowest set of locants; cite stereochemistry (E/Z, R/S) where needed.
6. With several groups, the senior one is the suffix/parent; the rest are prefixes.